

PROJECT FINAL REPORT GUIDE

Smart Tourism System

1 Report Structure

1.1 Title Page

- **Project Title**
- **Class**
- **Group ID**

1.2 Team Members & Task Allocation

Provide a clear table of satisfied team members and their specific contributions. **Every member must contribute.**

ID	Full Name	Role	Specific Tasks Assigned
1234567	Nguyen Van A	Leader, Backend	Designed architecture, implemented recommendation algorithm
1234568	Tran Thi B	Data Engineer	Collected restaurant data, cleaned datasets
1234569	Le Van C	AI Engineer	Implemented Sentiment Analysis model on reviews
1234570	Pham Thi D	Frontend Dev	Built Streamlit web app interface
...	...	...	...

1.3 Problem Analysis

- **Context:** Briefly describe the problem
- **Goal:** What does your system achieve?
- **Use Case Scenarios:** Describe key scenarios.

1.4 Computational Thinking Application

Demonstrate how you applied the logical steps of Computational Thinking.

1.4.1 Decomposition & Pattern Recognition

- **Decomposition:** Break down the big problem into smaller modules (Data Module, AI Module, User Interface, Recommendation Engine).
- **Pattern Recognition:** What similarities did you find? (e.g., “Users usually care about Price and Distance first”, or “Restaurant data has consistent fields”).

1.4.2 Abstraction

- What data did you keep and what did you ignore?
 - **Key Attributes:** e.g., `Restaurant(Name, Location, Price, Rating, Cuisine_Tag)`.
 - **Ignored:** Owner’s phone number, establishment date, etc.
- Explain *why* these attributes are sufficient for the AI/Algorithm.

1.4.3 Algorithm Design

- **Flowchart:** Include a diagram of your system logic.
- **Pseudocode:** Write out the core logic of your recommendation or processing function.

1.5 Implementation

This section details the technical execution.

1.5.1 Data Collection

- **Source:** Where did you get the data?
 - *Open Data:* Google Maps API, OpenStreetMap, VNAT.
 - *Scraped:* Agoda, Tripadvisor, Foody.
 - *Datasets:* Kaggle (e.g., Hanoi Food Review dataset).
- **Description:** Size of dataset, features collected.

1.5.2 AI Models

Describe the AI/ML components used.

- **Recommendation System:** Content-Based Filtering, Collaborative Filtering, or Knowledge-Based.
- **NLP:** Chatbot (RAG/LLM), Sentiment Analysis on reviews, Machine Translation.

- **Computer Vision:** Landmark recognition, Menu translation/OCR.
- **Tools:** scikit-learn, tensorflow, huggingface, openai API.

1.5.3 Web Application Development

- **Frameworks**
- **Screenshots:** Include key screenshots of the running app.

1.6 Testing & Improvement

- **Testing:** Show examples of inputs and resulting outputs.
- **Evaluation:** Does the AI give good results?
- **Limitations:** What is missing? (e.g., Real-time traffic not integrated).
- **Future Improvements:** What would you add next?

1.7 Conclusion

- Restate the value of the project.
- Lessons learned about Computational Thinking and AI.

1.8 References

- List APIs, Datasets, Libraries, and Tutorials used.

2 Deliverables

All materials should be uploaded to your cloud folder (Google Drive / OneDrive).

1. **Report:** [Group_ID]-report.pdf
 - **Page Limit:** ~30 pages (excluding cover page and references).
 - **Word Count:** ~4000 words.
2. **Slides:** [Group_ID]-slide.pptx or [Group_ID]-slide.pdf
3. **Source Code:** src folder (containing all code).